

Lecture 3

Bargaining

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Start from the ultimatum game

Let x denote the share of a payoff (normalized to one) that goes to Pl. 2

Pl. 1 offers a share x to Pl. 2

If Pl. 2 accepts, then Pl. 1 and Pl. 2 respectively receive $1 - x$ and x

If 2 rejects the offer, both get zero

The stark prediction follows from 2's weakness in this game: no "bargaining power"

Let's allow for counter-offers

Assume Pl. i discounts future payoffs at rate δ_i

Pl. 1 offers a share x to Pl. 2

If Pl. 2 accepts, then Pl. 1 and Pl. 2 respectively receive $1 - x$ and x

If 2 rejects the offer, then Pl. 2 gets to make a proposal; if 1 accepts, they each get their own share. If Pl. 1 rejects, she gets to make a proposal in the following period...

...and so on for T (even) periods

At the end of period T , all surplus gets destroyed

We call this, the alternating offers game (AOG_T)

What happens?

Proposition

Proposition 1. *There is a unique SPE, in which Pl. 1 in the first round offers*

$$x_1^* = 1 - (1 - \delta_2) \frac{1 - (\delta_1 \delta_2)^{\frac{T}{2}}}{1 - \delta_1 \delta_2},$$

which gets accepted

Sketch of the Proof

At $t = T$, Pl. 2 proposes $x_T = 1$

At $t = T - 1$, Pl. 1 proposes $x_{T-1} = \delta_2$

At $t = T - 2$, Pl. 2 proposes $x_{T-2} = 1 - \delta_1(1 - \delta_2)$

At $t = T - 3$, Pl. 1 proposes $x_{T-3} = \delta_2(1 - \delta_1(1 - \delta_2))$

At $T = 1$, Pl. 1 proposes $x_1 = \delta_2(1 - \delta_1(1 - \delta_2(\dots)))$

$$\begin{aligned}x_1 &= \delta_2 - \delta_2\delta_1 + \delta_2^2\delta_1 - \delta_2^2\delta_1^2 + \dots - \delta_2^{\frac{T}{2}-1}\delta_1^{\frac{T}{2}-1} + \delta_2^{\frac{T}{2}}\delta_1^{\frac{T}{2}-1} \\&= 1 - 1 + \delta_2 - \delta_2\delta_1 + \delta_2^2\delta_1 - \delta_2^2\delta_1^2 + \dots - \delta_2^{\frac{T}{2}-1}\delta_1^{\frac{T}{2}-1} + \delta_2^{\frac{T}{2}}\delta_1^{\frac{T}{2}-1} \\&= 1 - (1 - \delta_2)(1 + \delta_2\delta_1 + \dots + \delta_2^{\frac{T}{2}-1}\delta_1^{\frac{T}{2}-1}) \\&= 1 - (1 - \delta_2) \sum_{t=1}^{\frac{T}{2}} (\delta_2\delta_1)^{t-1} = 1 - (1 - \delta_2) \frac{1 - (\delta_1\delta_2)^{\frac{T}{2}}}{1 - \delta_1\delta_2}\end{aligned}$$

Exercise

What happens if T is odd?

The infinite horizon case

Rubinstein (1982) studies this problem

Proposition

Proposition 2. *AOG_∞ has a unique SPE: at any period in which Pl. i makes a proposal, she offers to Pl. j*

$$\delta_j \frac{1 - \delta_i}{1 - \delta_i \delta_j},$$

and so player i 's proposed share is

$$\frac{1 - \delta_j}{1 - \delta_i \delta_j},$$

and Pl. j accepts any offer weakly above it, rejects any lower offer

Proof

Let $\underline{v}_i$ (resp. $\bar{v}_i$) be the lowest (resp. the highest) payoff i receives in any SPE in a subgame in which she makes the initial offer

Consider a subgame in which she makes the initial offer; j will not accept anything below $\delta_j \underline{v}_j$, so $\bar{v}_i \leq 1 - \delta_j \underline{v}_j$

Similarly, j will accept anything that gives her more than $\delta_j \bar{v}_j$, so individual rationality requires $\underline{v}_i \geq 1 - \delta_j \bar{v}_j$

The same holds for j , so we must have

$$\bar{v}_i \leq 1 - \delta_j \underline{v}_j$$

$$\underline{v}_i \geq 1 - \delta_j \bar{v}_j$$

$$\bar{v}_j \leq 1 - \delta_i \underline{v}_i$$

$$\underline{v}_j \geq 1 - \delta_i \bar{v}_i$$

Proof (cont'd)

Combining the first two equations yields

$$\bar{v}_i - \underline{v}_i \leq \delta_j(\bar{v}_j - \underline{v}_j)$$

$$\bar{v}_j - \underline{v}_j \leq \delta_i(\bar{v}_i - \underline{v}_i)$$

which gives

$$\bar{v}_i - \underline{v}_i \leq \delta_j \delta_i (\bar{v}_i - \underline{v}_i) \Rightarrow \bar{v}_i = \underline{v}_i \equiv v_i$$

Proceeding analogously for j , we obtain a unique pair of SPE payoffs (v_i, v_j) ; plugging them into (1)-(4) yields

$$v_i = 1 - \delta_j v_j$$

$$v_j = 1 - \delta_i v_i$$

The system has a unique solution, s.t. $1 - v_i = \delta_j v_j$: j is indifferent between accepting and rejecting $\square$

Properties

Efficiency (no-delay) [why is delay inefficient?]

Payoff increases in own patience

First-mover advantage:¹ $v_1^* - v_2^*|_{\delta_1=\delta_2} > 0$

1. Decreasing in frequency of proposals (higher δ), it disappears as $\delta \rightarrow 1$

Exercise: Risk of breakdown

Suppose that after rejection, with probability $p > 0$ negotiation terminates, with payoffs (b_1, b_2)

For simplicity, no discounting ($\delta_1 = \delta_2 \approx 1$)

Claim 1. *The game has a unique SPE*

$$v_1 = \frac{1 - b_2 + (1 - p)b_1}{2 - p}$$

$$v_2 = \frac{1 - b_1 + (1 - p)b_2}{2 - p}$$

prove this claim

Legislative Bargaining: Baron-Ferejohn (1989)

Legislative bargaining among n legislators

Allocate a budget of size 1

Two variants of bargaining game:

1. Closed rule (no amendments)
2. Open Rule (in the paper, not covered)

Legislative session of

1. fixed length (if not bill passed, everyone gets zero)
2. infinite length: ends when bill passed

Preferences

Each legislator cares only about her allocation

- distributive politics application

Only receives payoff once bill is passed

Discount factor is δ , so indifferent between b_i in period t and $\delta^{t-1}b_i$ today

How Game is Played: Closed Rule

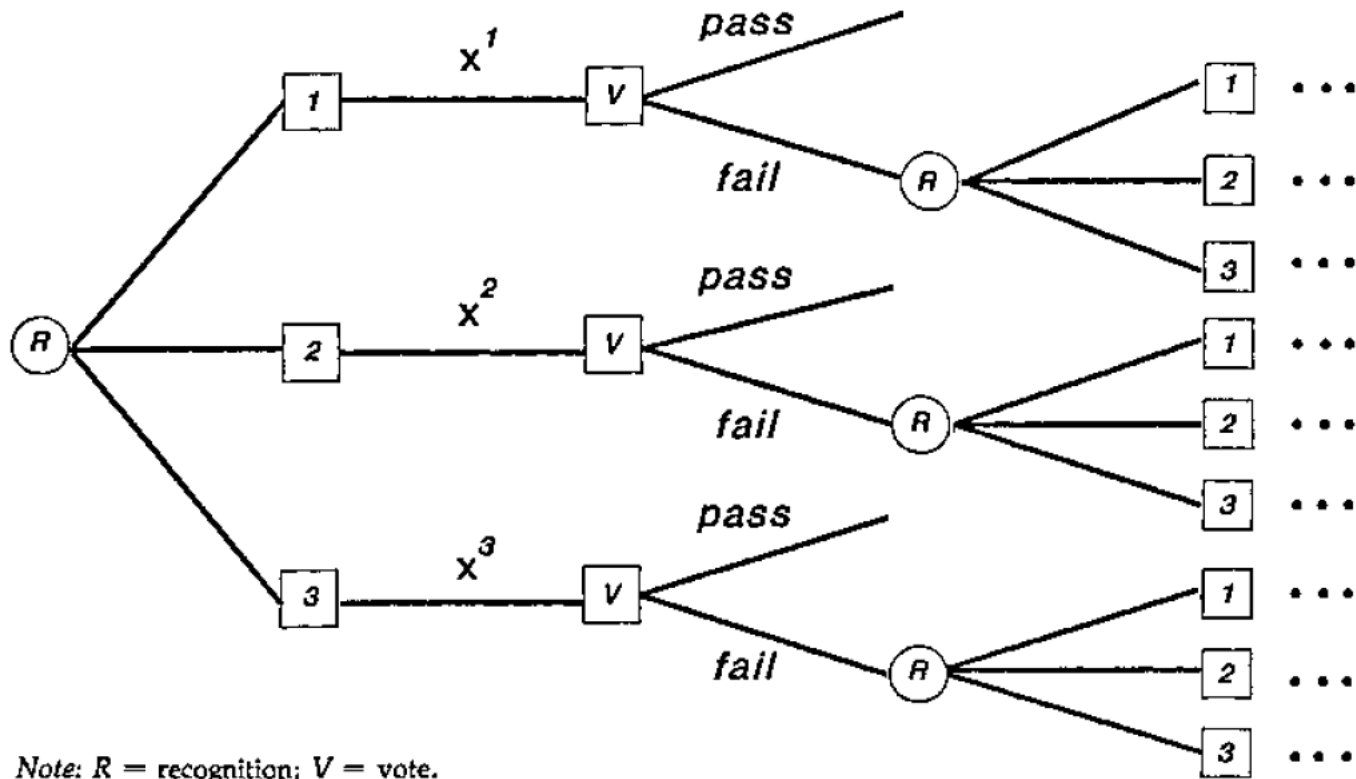
In each round t , each legislator i is *recognized* as proposer with probability p_i (often $p_i = \frac{1}{n} \forall i \in N$)

Proposer proposes a division of the budget x^t

Majority rule vote on proposal

- If pass, game ends and budget divided according to $x^t \in \Delta^{n-1}$
- If fail, game continues with new recognition round

Legislative Process: Closed Rule



[NOTA BENE: he uses x^i to denote i 's proposal]

2-Period Closed Rule

Assume $p_i = \frac{1}{n}$ and n odd

Second period, vote yes on any proposal, so proposer keeps full budget

Legislator i votes "yes" for proposal in first period if and only if $x_i^1 \geq \delta p_i = \frac{\delta}{n}$

First period proposer offers exactly $\frac{n-1}{2}$ other legislators $\frac{\delta}{n}$ and zero to everyone else

Proposer keeps $1 - \frac{\delta(n-1)}{2n}$ for herself

Bill passed with certainty in first round

2-Period Closed Rule Generalized

Suppose p_i different for different legislators

Still vote yes for any proposal in second round

Legislator i votes "yes" for proposal in first period if and only if $x_i^1 \geq \delta p_i$

First period proposer offers δp_i to the $\frac{n-1}{2}$ legislators with the **lowest** p_i

Weaker members have highest *ex ante* values of the game

Things to Notice

Proposer is very advantaged

When $p = \frac{1}{n}$ expected equilibrium payoff ($=v_i^*$) is the same for any finite-period

Equilibrium is efficient because first period proposal always passed (no delay)

v_i^* share is not always monotonic in proposal rights (why?)

Infinite number of periods: Punishment Strategies

Proposition

Proposition 3. *For an n -member, majority rule legislature with an infinite number of periods and a closed rule, if $1 > \delta > \frac{n+2}{2(n-1)}$ and $n \geq 5$, any allocation x may be supported in a SPE*

- x proposed by any member recognized and voted for by everyone
- If $y \neq x$ is proposed, majority $M(y)$ reject y . Next member proposes $z(y)$ with 0 to deviator
- If next member fails to propose $z(y)$, reject and next member punishes the member who didn't punish

Punishment, cont'd

- Anything is an equilibrium
- These equilibria rely on complicated punishment strategies
- How plausible are such state-contingent strategies?
- Select an equilibrium in which **players only condition on payoff-relevant aspects of the history**

Stationarity

Two subgames are **structurally equivalent** if

1. Bills being considered are the same
2. Sets of legislators eligible to be recognized next are the same (open rule only)
3. Legislators' strategy sets are the same (open rule only)

An equilibrium is **stationary** if all agents have same continuation values in all structurally equivalent subgames

- If two situations are the same with respect future payoffs, then players exhibit same behavior

Stationary Equilibrium: 3 Player, Symmetric Example

Suppose i is recognized. Proposes (x_1, x_2, x_3)

Wants to maximize x_i while getting bill passed

Suppose continuation value (i.e., expected payoff before recognition) for generic player j is v_j

Proposer gives one other player, chosen by the flip of a coin, exactly her continuation value δv_j

What do Proposals Look Like

$$v_j = \frac{1}{3}(1 - \delta v_j) + \frac{1}{3}\delta v_j \quad \Rightarrow \quad v_j = \frac{1}{3}$$

So proposer keeps $1 - \frac{\delta}{3}$ and gives one other player $\frac{\delta}{3}$

All players must have equal probability of being second coalition member. If not, least likely has lowest continuation value, so everyone should choose her. Which is a contradiction (why?)

Exercise: show that these continuation values can also be supported by a *pure strategy profile*

The General Result

Proposition

Proposition 4. *For all $\delta \in [0, 1]$ a configuration of pure strategies is a stationary subgame perfect NE in an infinite session, symmetric ($p = \frac{1}{n}$), majority rule, n -member (with n odd) legislature under closed rule if and only if it has the following form:*

1. *A member recognized proposes to receive $1 - \delta \frac{n-1}{2n}$ and offers $\frac{\delta}{n}$ to a coalition of $\frac{n-1}{2}$ other members*
2. *Each member votes for any proposal in which she receives at least $\frac{\delta}{n}$*
3. *Each member has an equal probability of being included in the coalition*

In this equilibrium, the bill passes in the first round

For uniqueness, see Eraslan (2002), *JET*

Now, if you are curious, Open Rule

How Game is Played: Open Rule

Each round, legislator recognized, proposes a division

Then a second player is recognized and may either

- Move the previous question
- Offer an amendment in the nature of a substitute

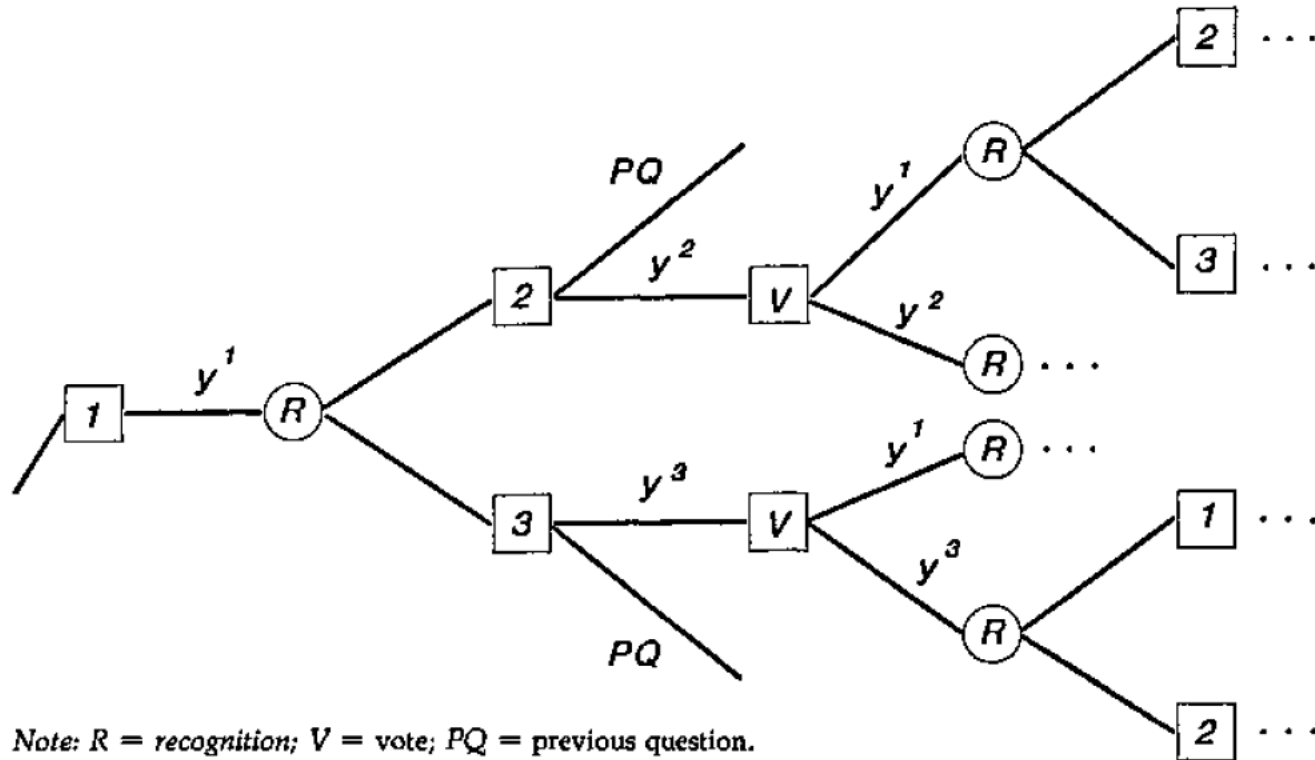
If previous question moved, vote on original proposal

- If pass, game ends, budget divided as per proposal
- If fail, game continues with new recognition

If amendment offered voted against proposal on floor

- If amendment wins, it becomes question on floor and a potential amender is recognized
- If amendment loses, bill remains question on floor and a new potential amender is recognized

Legislative Process: Open Rule



The Basic Logic of the Open Rule

Proposer needs someone to move the previous question and then to win final vote

- “Buy” a coalition willing to move the question and vote for bill
- Hope one of your coalition members is the amender
- May want to buy supermajority

Members of coalition must be indifferent between moving the previous question and offering most preferred amendment

Distribution of benefits more equitable, but don't get social efficiency

Setting

Simplified version from Cho and Duggan (2003), for a more general version see Banks and Duggan (2000)

Rather than splitting a dollar, group chooses a **policy** $x \in \mathbb{R}$

Players' policy preferences represented by the following utility function

$$u_i(p) = -(x - x_i)^2$$

Suppose n players are ordered in terms of their ideal point

Slightly different bargaining protocol compared to BF

Bargaining protocol

At every period, Pl. i is recognized with probability ρ_i

Makes a proposal $p_i \in \mathbb{R}$

- If a majority approves, $x = p_i$ and p_i remains in place forever
- If a majority rejects, a status quo $q \in \mathbb{R}$ is implemented, and next period a new proposer is drawn

Expected payoff if policy x is implemented at t

$$(1 - \delta^{t-1})u_i(q) + \delta^t u_i(x)$$

Fact: the bargaining protocol is strategically equivalent to the BF structure up to a linear transformation in payoffs

Key differences

Policy affinity shapes coalitions (sharper predictions)

Proposal rights create ‘policy externalities’

Technical issues in establishing that the median is decisive
(we will look at these issues in detail in a few weeks)

Typically, two sets of legislators:

- Those close enough to the median get to pass their ideal policy
- Those too far have to compromise

“U-shaped” equilibrium distribution of policies